Working assumptions, not forecasts. Capacity figures are scaled from a single Dutch reference case. Legal and regulatory entries were researched in September 2026, are not yet verified against primary sources, and will date. Generated 2026-09-03. How this was built.

Luxembourg

Sovereign government data centre network — capacity, legal posture, provider landscape and migration path.

1 Starting point

Population	0.68 m
GDP	EUR 90 bn
Public administration (NACE O)	38 k
Electricity price	171.7 EUR/MWh
Renewables	20.5%
Live hyperscaler regions	0

2 What is structurally different

- Micro-state geography. The Dutch rule of 3-5 regions with 50-100 km separation cannot be met inside the national territory. The model therefore assumes two in-country sites carrying 55/45 of the load and no in-country reserve. A credible disaster-recovery posture requires an out-of-country partner site - deferred to the federation layer.
- No hyperscaler region in-country. Unlike the Netherlands, there is no commercial hyperscale tier to fall back on for non-critical workloads without leaving the jurisdiction. The sovereign core therefore has to be sized for a larger share of total government demand, and the hybrid model (Dutch GOAL.md section 7) needs a cross-border commercial tier.

3 Capacity

Servers	526 (CPU 354 / GPU 27 / storage 145)
IT critical load	0.9 MW
Facility design load	1.3 MW
Sites	2 (by capacity 1, floor 2)
Average per site	0.7 MW
Site count set by	the minimum-sites floor, not capacity
CAPEX	EUR 32 m
OPEX	EUR 3 m / yr

4 Proposed geography

Region	Role	Share	MW	Notes
Luxembourg City - Bettembourg	Primary civil cloud	55%	0.7	CTIE and LuxConnect Tier IV estate, LU-CIX.

Region	Role	Share	MW	Notes
Bissen / North	Sovereign secondary	45%	0.6	30 km separation is the national maximum; Google Bissen site proves grid.

First-pass geographic hypotheses encoding only the obvious constraints, to be replaced by scored site selection.

5 Legal and regulatory posture

Governing instrument	Government cloud policy operated by CTIE; hosts the Estonian data embassy under treaty
Cloud certification	No national scheme; ISO 27001
Data classification	National classified-information law aligned to EU/NATO markings
Procurement route	Centre des technologies de l'information de l'Etat (CTIE)

Foreign jurisdiction exposure. Google Cloud region in Luxembourg; state workloads largely on CTIE infrastructure

Under the US CLOUD Act and FISA 702 a provider subject to US jurisdiction can face a lawful order for data it holds regardless of where that data sits. Residency is necessary but not sufficient; what matters is who holds the keys and who can be compelled.

6 Current state and provider landscape

Government cloud	Clarence (Proximus + LuxConnect, Google Distributed Cloud disconnected, 2024); CTIE GovCloud; LuxConnect state-owned Tier IV DCs; MeluXina HPC; hosts Estonia's Data Embassy
Maturity	operational
Digital identity	LuxTrust / eID / GouvID
Interconnection	LU-CIX; landlocked (Teralink to FRA/PAR/AMS/BRU)

7 Migration path and cost

Phase	Scope	MW	CAPEX	Cumulative	Hybrid
1	Sovereign core	0.3	EUR 5 m	16%	no
2	Security and defense	0.5	EUR 12 m	55%	no
3	State record	0.2	EUR 6 m	73%	no
4	Elective	0.3	EUR 9 m	100%	no

Phase 1 is the number that matters: **EUR 5 m** for 0.3 MW, 16% of total CAPEX. That is the floor below which no hybrid arrangement helps — and it is a small fraction of the full build.

8 Geography and threat notes

Very low seismic/flood; scarce land and strict zoning; ~80%+ of electricity imported; small labour pool; very low geopolitical risk; strong existing Tier-IV base